

Biology Syllabus

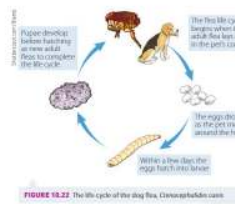
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Module 7: Infectious Disease

Causes of Infectious Disease

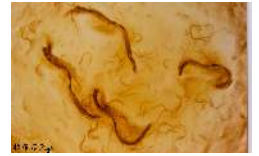
Inquiry question: How are diseases transmitted?

Syllabus dot point	Summaries
- describe a variety of infectious diseases caused by pathogens, including microorganisms, macroorganisms and non-cellular pathogens, and collect primary and secondary-sourced data and information relating to disease transmission, including: (ACSBL097, ACSBL098, ACSBL116, ACSBL117) - classifying different pathogens that cause disease in plants and animals (ACSBL117)	The immune system is a complex network of responses to outside threats. Infectious diseases involve threats from infectious agents known as pathogens. There is a fine balance between an appropriate and useful response and one that may harm the host organism. Disease results when there is a loss of balance between defence and attack. FIGURE 10.4 Broad classification of pathogens FIGURE 10.5 Cellular and non-cellular pathogens, and examples of diseases they cause (CJD: Creutzfeldt-Jacob disease) Cellular Pathogens - Parasites - Macroscopic - Eukaryotic multicellular organisms - Feed off the host body - Endoparasites live inside the host's body (e.g. helminths) - Ectoparasites live outside and feed from the host's body (leech) Symbol Plants Animals Agar Plate



Dwarf mistletoe weakens host trees by slowly robbing them of nutrients and water

Roundworms cause vomiting, weight loss, food stoppage and are mainly in kittens and puppies



2. Protozoa

- Microscopic
- Eukaryotic single-celled organisms
- Heterotrophic and cause damage by stealing nutrients, reproducing rapidly to form colonies and secreting chemicals that affect the host* body

Symbol



Plants

Sudden wilt disease causes sudden wilt in roots and stalks mainly with the phloem and spear base.

Animals

Coccidiosis is the destruction of intestinal tissues in dogs, that show no sign of illness

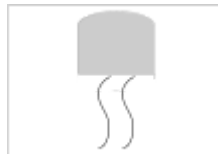
Agar Plate



3. Fungi

- Microscopic
- Eukaryotic multicellular organisms
- Have a cell wall (so do plants but they have chloroplasts)
- Heterotrophic + causes damage by secreting digestive enzymes to get food and by producing spores that release harmful enzymes

Symbol



Plants

Anthraxnose causes dark lesions on leaves

Animals

Ringworm is highly contagious and causes irritation

Agar Plate



4. Bacteria

- Microscopic
- Prokaryotic unicellular organisms
- Heterotrophic or autotrophic (hurt us through excess waste)
- Reproduce by binary fission and can very quickly grow into huge numbers under the right conditions
- They secrete toxins that disrupt cell function, sometimes deliberately attacking and invading our cells (for food)
- Decomposers

Symbol

Plant

Vascular wilts are extremely destructive diseases that cause wilting in plants

Animal

Botulism is accompanied by paralysis and other fatal consequences

Agar Plate



Non-cellular Pathogens

5. Virus

- Microscopic
- Non-nuclear agents - made of DNA or RNA enclosed in a protein coat
- Reproduce through hijacking living cells and causing the infected cell to produce more until it explodes - sending virus particles everywhere
- Viruses have two choices
 - Reproduce more quickly → become less deadly
 - Be more deadly → way harder to reproduce

Symbol



Plant

Abutilon mosaic virus creates an attractable mosaic pattern that causes vein-limited blotches

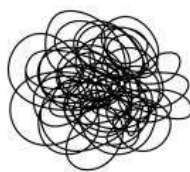
Animal

COVID-19 causes excessive coughing, flu-like symptoms, and blockage in lungs that may be fatal.

6. Prions

- Microscopic
- Non-nuclear agents made of incorrectly folded proteins (proteinaceous)
- Reproduce by transmitting the misfolded protein to other cellular proteins

Symbol



Plant
N/A

Animal

Chronic wasting disease causes behavioural changes, lack of coordination and eating difficulties that reside in animals.

- investigating the transmission of a disease during an epidemic

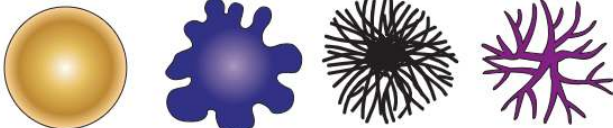
Factors:

1. A pathogen
2. A susceptible host
3. A favourable environment to reproduce or replicate
4. Transmission between affected and unaffected individuals

Cause	Equine Influenza (Horse Flu Australia 2007)
Transmission	- Nasal discharges

		- Infected horses cough secretes nasal discharges inhaled by unaffected horses - Farm equipment - Human carriers (skin, hair, clothing, shoes)
	Method	- horse inhales virus → pathogen replicats in upper respiratory - when number of pathogens is enough, virus will break through cells - this causes nasal discharges to other horses and farming equipment
	Host Reponse	- fever - nasal discharge - cough - lethargy - loss of appetite - weakness
	Prevention & Control	- keep horses up-to-date with vaccinations - continuous monitoring of virus - avoid sharing equipment (hinders transmission) - quarantine operations to isolate - anti-inflammatory drugs
	Control of epidemic	- affected horses isolated - quarantine for 14 days - public education - biosecurity measures and training - analyse epidemic (identification of prone area and past outbreaks)

– design and conduct a practical investigation relating to the microbial testing of water or food samples	Microbes surround and are all around us as organisms. Detection of microbes can only be done through microscopes as it is not visible to the naked eye. Agar jelly in an incubator of 30 degrees is suitable for microbial growth. Equipment: - Food sample - 6x agar plates - 1x inoculating loop - 1x bunsen burner - Alcohol (for cleaning, like baby wipes) - 3x test tubes - Pen & Labels - Sticky tape Procedure: 1. Sterilise table surfaces using cleaning alcohol 2. Sterilise the inoculating loop by heating it under the Bunsen burner. Also, sterilise the three test tubes by heating the 'mouth' (opening) of the test tubes to restrict/minimise pathogens in the air entering the tubes 3. Transfer small quantities of three different food samples into three test tubes with 2mL of distilled water and gently mash up the food samples 4. Dip the sterilised inoculating loop into the food sample in one of the test tubes and wipe the food gently on the surface of the agar plates 5. Put on the agar plate lid and seal the plate using the sticky tape, then label the food type on the lid 6. Heat the inoculating loop again and repeat steps 4 & 5 for the two other types of food samples in the remaining test tubes 7. Leave 3 sealed agar plates as the control 8. Observe and record observations		
	Hazard	Risk	Prevention

	<table><tr><td>Bacteria</td><td>Infection</td><td>Wear gloves, keep the agar plate sealed, and wash hands thoroughly using water and disinfectant hand-wash solution</td><td>Contact teacher</td></tr><tr><td>Fire</td><td>Burn</td><td>Take care of using the Bunsen burner and hot inoculating loop</td><td>Run hands under water, contact the teacher</td></tr></table>	Bacteria	Infection	Wear gloves, keep the agar plate sealed, and wash hands thoroughly using water and disinfectant hand-wash solution	Contact teacher	Fire	Burn	Take care of using the Bunsen burner and hot inoculating loop	Run hands under water, contact the teacher
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	a Form  Circular Irregular Filamentous Rhizoid b Elevation  Raised Convex Flat Umbonate Crateriform c Margin  Entire Undulate Filiform Curled Lobate FIGURE 10.37 Different types of bacterial colonies can be identified, based on their a form, b elevation and c margin.								
– investigate modes of transmission of infectious diseases, including direct contact, indirect contact and vector transmission	Transmission → direct (body fluids) - Blood - Saliva - Sexual fluids → non-direct - Airborne - Objects → vector (carriers) - Mosquitoes - Fleas								
• investigate the work of Robert Koch and Louis Pasteur, to explain the causes and transmission of infectious diseases, including – Koch's postulates – Pasteur's experiments on microbial contamination	Robert Koch - Developed the agar-plate technique - Some diseases are caused by microorganisms - Criteria - The same microorganism must be present in every diseased host - Micro-organisms must be isolated and cultured in the laboratory with recording and accurate descriptions - When the pure culture sample is inoculated into a healthy host → the symptoms must be the same - The micro-organism must be isolated from the second host and identified the same as the original species Louis Pasteur - The process of pasteurisation was to kill disease-causing micro-organisms								

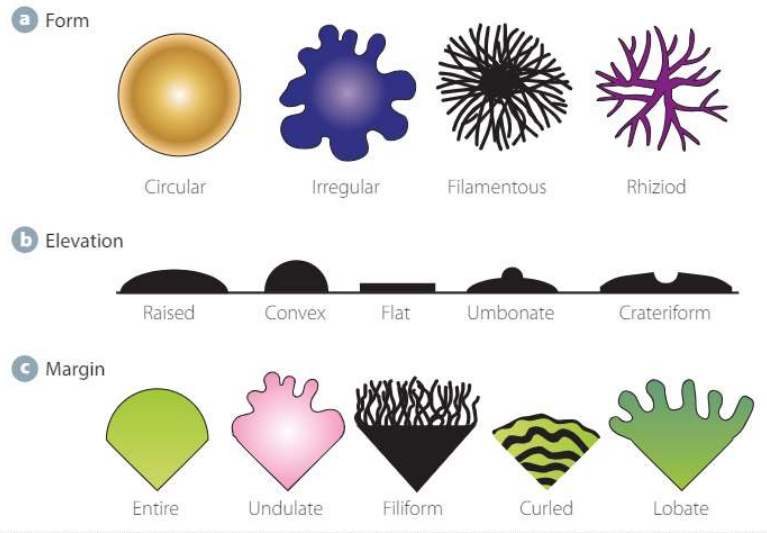


FIGURE 10.37
Different types of bacterial colonies can be identified, based on their **a** form, **b** elevation and **c** margin.

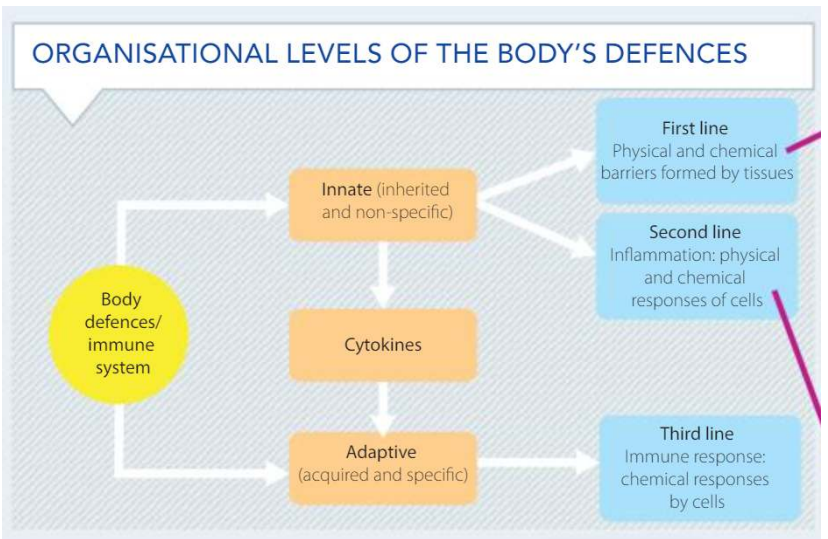
	- Pasteur's swan-necked flask disproved the theory of spontaneous generation due to the trapped bacteria - Pasteur created a variety of vaccines (cholera and rabies)		
● assess the causes and effects of diseases on agricultural production, including but not limited to: - plant diseases - animal diseases	Australia is isolated from the rest of the world, hence the risk of a disease in agriculture is less. Two types of plant and animal disease are areas of concern. - Endemic diseases: diseases consistent in a region E.g. bovine Johne's disease in cattle, sheep and goats - Exoytic (introduced) diseases E.g. foot and mouth disease, avian influenza diseases in agriculture are in three factors - Host factors - Susceptibility to disease, access to pathogen - Concurrent disease or poor nutrition - Pathogen factors - Pathogen's availability - Ability to transfer between hosts - Success establishment - Environmental factors - Overcrowding - Build up wastes - Favourable environment Factors contributing to the risk of infectious disease <table border="0"> <tr> <td> - Increased mobility of humans - Rise of intensive and industrial-type agriculture - Changing patterns of land use e.g. deforestation - Climate change - Antimicrobial resistance </td><td> - Pesticide resistance - Pesticide resistance - Loss of genetic diversity - Increase in 'hobby farmers' - Increase the use of aquaculture </td></tr> </table>	- Increased mobility of humans - Rise of intensive and industrial-type agriculture - Changing patterns of land use e.g. deforestation - Climate change - Antimicrobial resistance	- Pesticide resistance - Pesticide resistance - Loss of genetic diversity - Increase in 'hobby farmers' - Increase the use of aquaculture
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- plant diseases	→ plants Abscission (dropping of leaves) area - a plant response to pathogens - Fungi - Generally transmitted through wind, water, and contact with resoioouvors - Damage plants by conducting tissues and absorbing nutrients - Insect and mites - Damage to plant tissue - Direct vectors to pathogens - E.g. fruit fly - Bacteria - Occur in soil, weeds, and seeds - Harbour bacteria in hands and spreads - E.g. bacterial blight of peas - Nematodes - Live in soil but some act as a plant pathogen - Attacks roots causing galls and lumps - E.g. root-knot nematode - Viruses - Obligate intracellular parasites - E.g. tobacco mosaic viruses - Phytoplasmas - Related to bacteria (don't have a cell wall" - Transmitted by insect vectors and inhabit phloem tissue Effects:		

	- Biological effects on plants (death, abnormal growth) - Social and economic on farmers (reduced trading, losses, reduced yields) - Social and economic economy
- animal diseases	→ animals - Bacteria: tuberculosis - Virus: coronavirus - Fungus: ringworm - Protozoa: coccidiosis - Prion: chronic wasting disease - Macroparasite: roundworm Effects: - Death - Loss of appetite or weight - Economic loss - Low growth - Low fertility
• compare the adaptations of different pathogens that facilitate their entry into and transmission between hosts (ACSBL118)	Pathogens must - Enter the host - Multiply host tissues - resist/ not stimulate defence mechanisms - Damage the host → virus - Trojan horse - Enters host and adheres protein which tricks the cell into thinking it is food, activating endocytosis - After, it uncoats and fuses with cell DNA → bacteria - Iron man - Forms a biofilm that resists phagocytosis from the host cell → parasite - Alarms off - Secretes fluids that reduces host cell immune responses → prion - Ninja - Secretes factors for invasion - 'Piggyback' other proteins to move into the host → protozoan - Camouflage - Lysosomes fuse with cell membrane - Protoisium (to hide) → fungi - Batman - Fungi hooks on and releases enzymes to 'burn' through Adaptations for survival → airborne pathogens - Survive acid by turning urea into ammonia and hiding inside - Make you sneeze - Lightweight → oral faecal route pathogens - Survive acid and low oxygen

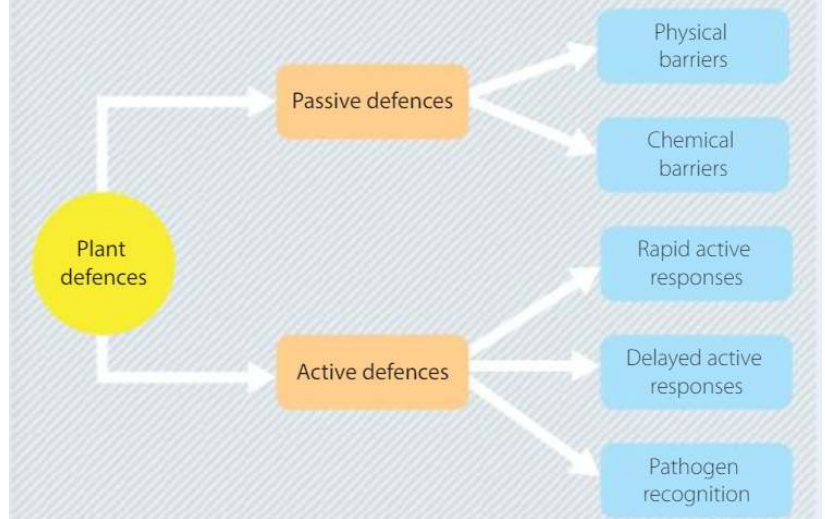
	- Diahorrea to help spread (diarrhea is like sneezing but for your butt) → Vector-assisted pathogens - Survive vector's body - The timing of reproduction matches the vector's feeding time → sexually transmitted pathogens - Jump from one fluid to another - Enter uterus and placenta - Mother → child
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Responses to Pathogens

Inquiry question: How does a plant or animal respond to infection?

Syllabus dot point	Summaries
● investigate the response of a named Australian plant to a named pathogen through practical and/or secondary-sourced investigation, for example: - fungal pathogens - viral pathogens	An introduced species of water mould that thrives in Australian conditions → Pathogen: It is a fungus that affects the plant <i>Eucalyptus marginate</i> causing the disease Phytophthora Dieback - It travels through the soil via zoospores - Must survive through wet conditions - When living conditions aren't ideal the thick chlamydospore walls protect the spores when the conditions are unfavourable and dry → Symptoms: - It attaches to the roots of the plant and absorbs its nutrients, causing the tree not to be able to absorb water or its nutrients from the soil - Slowly dying foliage → Response: - The plant has an increase in lignin production to protect the cell wall of plant cells as it is impervious (cannot pass through) to the pathogen
● analyse responses to the presence of pathogens by assessing the physical and chemical changes that occur in the host animals' cells and tissues (ACSB119, ACSBL120, ACSBL121, ACSBL122)	 The diagram, titled "ORGANISATIONAL LEVELS OF THE BODY'S DEFENCES", illustrates the hierarchy of the immune system. It begins with a yellow circle labeled "Body defences/ immune system". Two arrows lead from this circle to two orange boxes: "Innate (inherited and non-specific)" and "Adaptive (acquired and specific)". An arrow connects the "Innate" box to the "Adaptive" box, with the label "Cytokines" placed between them. From the "Innate" box, two arrows point to two light blue boxes: "First line: Physical and chemical barriers formed by tissues" and "Second line: Inflammation: physical and chemical responses of cells". From the "Adaptive" box, an arrow points to a third light blue box: "Third line: Immune response: chemical responses by cells".

RESPONSES BY PLANTS AGAINST PATHOGENS



→ plants

- Passive
 - Physical
 - Thick cuticle/bark
 - Small stomata
 - Vertical hanging leaves
 - Cell wall
 - Chemical
 - Presence of chemical compounds = reduce bacteria and fungi
 - E.g. glucosides, saponins, enzymes
 - Chemicals receptors = pathogens associated molecular patterns (PAMP)
- Active
 - Pathogen recognition
 - Detect certain physical and chemical signals
 - Genes within cells regulate plant responses
 - Rapid active response (minutes into hours)
 - Recognition of a pathogen by proteins causing changes in permeability
 - The release of hydrogen peroxide kills microbes directly
 - Cell-wall apposition: surround and isolate it
 - Delayed active response (days)
 - Repair damages
 - System acquired resistance: remembering the pathogen

Immunity

Inquiry question: How does the human immune system respond to exposure to a pathogen?

Syllabus dot point	Summaries
● investigate and model the innate and adaptive immune systems in the human body (ACSBL119)	Animals (such as humans) have 3 lines of defence to combat entering the body It includes → first line (non-specific) - Physical barriers : such as skin

	- Chemical barriers : secretions the body produces to trap and kill germs E.g. tears, mucus, earwax, stomach acid, saliva, beneficial bacteria and fungi - Biological boundaries : natural entry points into the body that can close tightly to stop pathogens E.g. eyes, mouth, anus → second line (non-specific) - Non-specific set of defences that engage once a breach has been made and pathogens have entered - This mainly involves a process called inflammation and the use of cells called macrophages - Macrophage (big - eater) - Type of white blood cell - Engulfs or 'phagocytosis' pathogens - Inflammation - Blood rushes to a site of infection causing it to swell and turn reddish - This is the body sending lots of macrophages to the attack site and clotting factors to close the entry point - YouTube: "In a nutshell- immune system"
● explain how the immune system responds after primary exposure to a pathogen, including innate and acquired immunity	→ Innate Immunity A general, non-specific response to a pathogen, characterised by the first and second lines of defence. First line of defence: - The first line of defence is all about stopping pathogens from even entering the body. This includes barriers such as: - Physical: skin - Chemical: secretions the body creates that trap and kill germs, such as tears, mucus, ear wax, saliva, stomach acid, beneficial bacteria and fungi. - Biological barriers: natural entry points into the body that can close tightly to prevent the entry of pathogens, such as eyes, anus, and mouth. Second line of defence: - The second line of defence contains a set of defences that engage once a breach has been made and pathogens have entered the body. This mainly involves a process of inflammation, where the blood floods the site of infection, causing it to swell and become reddish. During this process, the body sends cells called macrophages that are contained within the blood, and are a specific type of white blood cell that performs phagocytosis on the pathogens. → Adaptive/acquired immunity A specific response to a pathogen is characterised by the third line of defence. Third line of defence: - The third line of defence is a specific response to a pathogen that consists of specialised cells that attack the pathogen when the first two lines have been breached. - B and T lymphocytes (or B and T cells) are responsible for the generation of the adaptive immune system response - Each lymphocyte has a specific set of receptors capable of recognising a specific pathogen. There are millions of combinations available to the immune system, and our bodies are equipped from birth as a product of our genes to battle certain pathogens that we have not yet encountered. - B lymphocytes are stored in the lymphoid tissue - When a specific antigen enters the body, a B lymphocyte will initiate its response and destroy the pathogen. Once this is done, the B lymphocyte becomes a memory cell, ready to fight if this pathogen were to ever enter the body again.

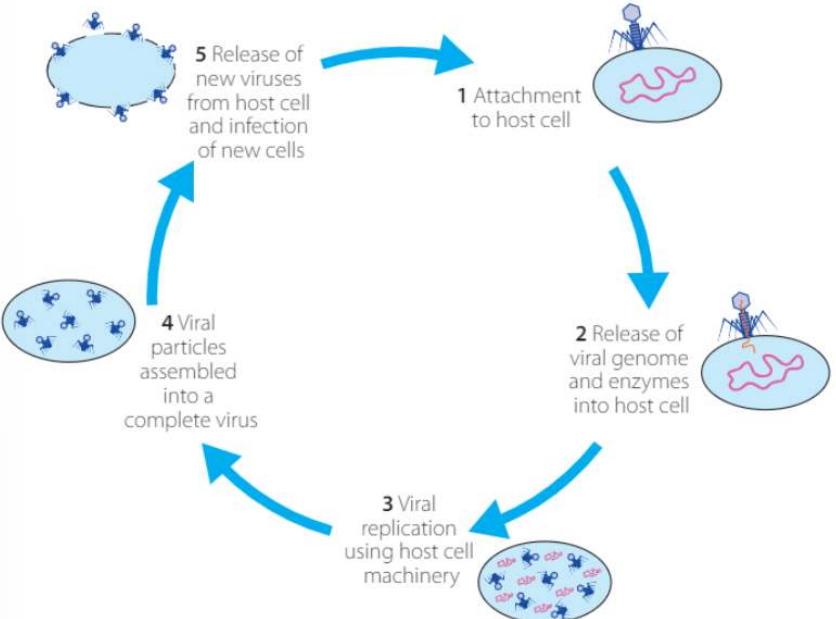
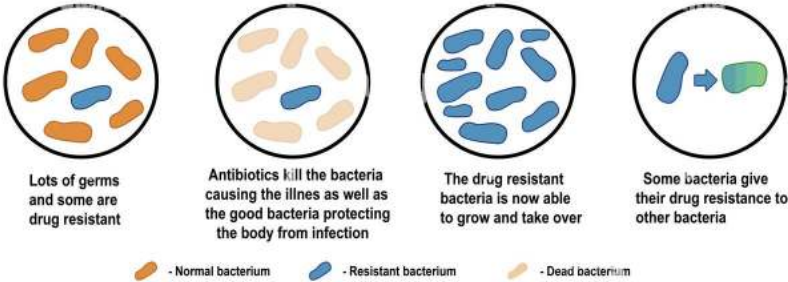
	- The third line of defence is so specific because it is only the B and T lymphocytes that recognise the specific pathogen that enters the body that will fight against it, rather than all of the lymphocytes. There are two types of adaptive immune system responses: The humoral response to pathogens is initiated after primary exposure to a pathogen outside the body cells. This response NEUTRALISES the pathogen - Helper T lymphocytes release substances that initiate a response from B lymphocytes, who upon reception make many copies of the specific B cell that can respond to the specific pathogen. These B cells are made into two different cell types, memory cells and plasma cells. - Antibodies/immunoglobulins are produced by plasma cells in response to exposure to a pathogen The cell-mediated immune system is a branch of the adaptive immune response that targets pathogens inside infected cells as well as tumour cells. This response DESTROYS the pathogen-infected cells - The main cell involved is the T lymphocyte, which develops in the bone marrow and matures in the thymus. - Mature T lymphocytes are present in the blood, spleen, tonsils, Peyer's patches and lymph nodes. - There are four main types of T lymphocytes.- Helper T cells activate the cloning of cytotoxic T cells as well as B cells.- Cytotoxic T cells bind with pathogen-infected cells and destroy them.- Memory T cells remain in the body to respond to future infections.- Suppressor T cells suppress the response once pathogens have been eliminated. → passive immunity When the body is exposed to immunoglobulins/antibodies to help the body prevent the disease from developing. During this process, no memory cells are produced.
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Prevention, Treatment and Control

Inquiry question: How can the spread of infectious diseases be controlled?

Syllabus dot point	Summaries
● investigate and analyse the wide range of interrelated factors involved in limiting local, regional and global spread of a named infectious disease	→ local : neighbourhood, town, village, city - sanitation, overcrowding and poor communication need to be managed → regional : five regions: Africa, Americas, Asia, Europe and Oceania - mountains, deserts, rainforests etc. → global - Pathogen factors - natural reservoirs, incubation period and virulence - Host factors - concurrent and malnutrition - Environmental and geographic - sanitation and environmental conditions - Other societal factors - mass human population, overcrowding, isolation
● investigate procedures that can be employed to prevent the spread of disease, including but not limited to: (ACSBL124) - hygiene practices	Hygiene - Personal : keeping body and openings clean E.g. washing hands, cleaning teeth, preventing airborne transmission - Community : prevent build-up in the community E.g. sewage and garbage disposal, sterilisation, reduce overcrowding - Clean food and water - Water quality aligns with standards

- quarantine	Quarantine - Geographic isolation - Increase trade and travel → creates more spread of disease - Minimise risk of external diseases entering - Animal - Quarantine stations - Checking regularly for signs of disease - Plant - Examining all types of plants - Treated by quarantine officers - Human - Human quarantine and isolation															
- vaccination, including passive and active immunity (ACSBL100, ACSBL123)	Vaccination - Introduction of a vaccine into the body - Immunisation : The body reacts to a vaccine by going through the immune response - the body produces memory cells for the antigen so that if it enters they can defend - Active acquired immunity - When immune response occurs and memory cells are produced - Can be naturally or artificially induced - For vaccines to be effective, booster shots must be given over a no of years - Passive acquired immunity - Introduction of antibodies in the body to prevent a disease															
- public health campaigns	Public health campaigns - Four categories - Resolution - governments and WHO to find solutions - Information - scientific studies - Coordination - efforts on a local regional and global scale - Education - affecting infectious disease transmission															
- use of pesticides	Pesticides Pesticides can be classified into three groups: - insecticides – kill insects - fungicides – kill fungal pathogens - herbicides – kill weeds. The problem is resistance and damaging effects															
- genetic engineering	Genetic engineering : altering the genetic composition of an organism - Disease-resistant plants and animals - Long-term ineffective															
● investigate and assess the effectiveness of pharmaceuticals as treatment strategies for the control of infectious disease, for example: - antivirals - antibiotics	<table border="1"><caption>TABLE 13.2 Classes of antimicrobial drugs</caption><thead><tr><th>CLASS OF MEDICATION</th><th>PATHOGENS TARGETED</th><th>EXAMPLES</th></tr></thead><tbody><tr><td>Antibiotics (antibacterials)</td><td>Bacteria</td><td>Penicillins, tetracyclines, polymyxins, sulfonamides, cephalosporins</td></tr><tr><td>Antivirals/ antiretrovirals</td><td>Viruses</td><td> • Tamiflu® (oseltamivir) used to treat influenza (antiviral) • Abacavir (antiretroviral used in HIV/AIDS treatment) </td></tr><tr><td>Antifungals</td><td>Fungi</td><td>Fluconazole, amphotericin B, caspofungin</td></tr><tr><td>Antiprotozoals</td><td>Protozoa</td><td>Doxycycline, metronidazole, mefloquine</td></tr></tbody></table> → antivirals : used to control viral infections - - Slow down its processes and allow the body's natural defences to take over	CLASS OF MEDICATION	PATHOGENS TARGETED	EXAMPLES	Antibiotics (antibacterials)	Bacteria	Penicillins, tetracyclines, polymyxins, sulfonamides, cephalosporins	Antivirals/ antiretrovirals	Viruses	• Tamiflu® (oseltamivir) used to treat influenza (antiviral) • Abacavir (antiretroviral used in HIV/AIDS treatment)	Antifungals	Fungi	Fluconazole, amphotericin B, caspofungin	Antiprotozoals	Protozoa	Doxycycline, metronidazole, mefloquine
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	 → antibiotics : control bacterial infections - slowing down or killing the growth of bacteria - however, antibiotic resistance can occur  - Normal bacterium - Resistant bacterium - Dead bacterium
investigate and evaluate environmental management and quarantine methods used to control an epidemic or pandemic	→ environment - Use of procedures and protocols - Provision of facilities including workspaces, hygiene maintenance and safe waste management (leak-proof bags, sterilisation) - Provision of personal maintenance including masks, gloves and sanitisation. → quarantine - Isolating patients in a single room with three metres apart - Disinfectants and cleaners throughout the day → long-term - Reducing transmission between humans-humans and humans-animals - Outbreak contamination measures
- interpret data relating to the incidence and prevalence of infectious disease in populations, for example: - mobility of individuals and the portion that is immune or immunised (ACSBL124, ACSBL125) - Malaria or Dengue Fever in South East Asia	→ incidence : no. of new cases occurring over a specified time - Can be the infection rate or probability of contracting the disease - Calculate $\frac{\text{no. of new cases over a specified time}}{\text{size of population at start of monitoring period}} \times 100$ → prevalence : no. of total (new and previous) cases during a time period - Calculate

	$\frac{\text{all new \& previous cases during a time period}}{\text{population during time period}} \times 100$ → mobility - A factor of potential outbreak - Rate of immunisation factor - Herd immunity : high number of vaccinated individuals to reduce unvaccinated from contracting the disease
● evaluate historical, culturally diverse and current strategies to predict and control the spread of disease (ACSB125)	- Historically, different strategies including quarantine have been used to prevent the spread of disease. - John Snow is known as the 'father of field epidemiology'. - Disease control is influenced by culture, and this needs to be taken into account when developing control measures.
● investigate the contemporary application of Aboriginal protocols in the development of particular medicines and biological materials in Australia and how recognition and protection of Indigenous cultural and intellectual property is important, for example: - bush medicine - smoke bush in Western Australia	- Traditional medicine, such as Aboriginal bush medicine, is a valuable source of knowledge that needs to be preserved for future generations. - The intellectual property rights of Indigenous peoples are not well supported in Australia as yet.